

UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

Maximum - Minimum I: Unconstrained Optimization

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0.1 Introduction

In our daily life we always deal with optimization problems such as problems related to the cost, revenue, profit, the best way to go to work, the best choice, the best shape etc,... The formulation or modelling of these phenomena of life can lead to very complicated mathematical problems.

The purpose of these lectures on maximum-minimum is not to solve real life problems but rather to present basic methods of solving simple optimization problems for functions of two or three variables.

Definition 0.1 (Point of local maximum/minimum - Critical point)

Let $A \subset \mathbb{R}^2$ and let $f : A \rightarrow \mathbb{R}$.

- (a) The point $(a, b) \in A$ is called a **point of local maximum** for the function f if there exists a small disc D centered at (a, b) such that

$$f(a, b) \geq f(x, y) \quad \forall (x, y) \in D \cap A.$$

- (b) The point $(a, b) \in A$ is called a **point of local minimum** for the function f if there exists a small disc D centered at (a, b) such that

$$f(a, b) \leq f(x, y) \quad \forall (x, y) \in D \cap A.$$

- (c) If f we assume that the function f is differentiable, then we say that (a, b) is a **critical point** for the function f if $\nabla f(a, b) = (0, 0)$ i.e., $f_x(a, b) = f_y(a, b) = 0$.

Definition 0.2 (Point of global maximum/minimum)

Let $A \subset \mathbb{R}^2$ and let $f : A \rightarrow \mathbb{R}$.

- (a) The point $(a, b) \in A$ is called a **point of global maximum** for the function f if

$$f(a, b) \geq f(x, y) \quad \forall (x, y) \in A.$$

- (b) The point $(a, b) \in A$ is called a **point of global minimum** for the function f if

$$f(a, b) \leq f(x, y) \quad \forall (x, y) \in A.$$

0.1.1 First order conditions for minimality/maximality

Theorem 0.3 (A necessary condition for local max/min) *Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a differentiable function. If the point (a, b) is a point of local maximum/minimum for the function f , then (a, b) is a critical point for f i.e., $f_x(a, b) = f_y(a, b) = 0$.*

Proof: Assume for instance that (a, b) is a point local maximum for the function f . Then $f(x, b) \leq f(a, b)$ for every number x in a small interval centered at a . This means that the function of a single variable $g(x) := f(x, b)$ has a local maximum at the number $x = a$. Since the partial derivative $f_x(a, b)$ exists, it follows that the function g is differentiable at $x = a$. So, we get from first year calculus that

$$g'(a) = 0 \quad \Leftrightarrow \quad f_x(a, b) = 0.$$

On the other hand, $f(a, y) \leq f(a, b)$ for every number y in a small interval centered at b . This means that the function of a single variable $h(y) := f(a, y)$ has a local maximum at the number $y = b$. Since the partial derivative $f_y(a, b)$ exists, it follows that the function h is differentiable at $y = b$. Therefore, we get from first year calculus that

$$h'(b) = 0 \quad \Leftrightarrow \quad f_y(a, b) = 0.$$

Hence, we finally obtain that $\boxed{f_x(a, b) = f_y(a, b) = 0 \iff \nabla f(a, b) = (0, 0)}$.

We follow the same steps in the case where (a, b) is a point of local minimum (which is left as an exercise). This completes the proof of the theorem. ■

Remark 0.4 **Theorem 0.3 suggests that the points of local maximum/minimum for a differentiable function should be selected from the pool of its critical points.**

0.1.2 Second order conditions for minimality/maximality

In order to find points of local maximum/minimum for a differentiable function f , we first find all its critical points. Assume that (a, b) is a critical points for the differentiable function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, i.e., $f_x(a, b) = f_y(a, b) = 0$.

Now, in order to decide on the nature of the critical point (a, b) i.e., to say whether (a, b) is a point of local maximum/minimum or neither, we will need some additional conditions. To this aim, we write

$$f(x, y) = f(a, b) + \mathcal{R}((x, y), (a, b))$$

and the remaining task will be to study the sign of the quantity $\mathcal{R}((x, y), (a, b))$.

For instance, if $\mathcal{R}((x, y), (a, b)) > 0$ for every (x, y) in a small disc centered at the point (a, b) then, (a, b) is a point of local minimum for the function f .

Now, looking at the second order Taylor formula for functions of two variables (studied in chapter on Differentiability-II), we assume that the function f is class $\mathcal{C}^2$, i.e., with continuous second order partial derivatives, and we write

$$\begin{aligned} f(x, y) &= f(a, b) + \frac{1}{1!} \nabla f(a, b) \cdot (x - a, y - b) + \frac{1}{2!} (x - a, y - b) \cdot Hf(\xi, \eta) \begin{pmatrix} x - a \\ y - b \end{pmatrix} \\ &= \boxed{f(a, b) + \frac{1}{2!} (x - a, y - b) \cdot Hf(\xi, \eta) \begin{pmatrix} x - a \\ y - b \end{pmatrix}}. \end{aligned}$$

where the point (ξ, η) lies on the segment line joining (a, b) to (x, y) .

So, the natural candidate as $\mathcal{R}((x, y), (a, b))$ is

$$\mathcal{R}((x, y), (a, b)) = \frac{1}{2!}(x - a, y - b) \cdot Hf(\xi, \eta) \begin{pmatrix} x - a \\ y - b \end{pmatrix}.$$

Please read the lecture notes on Differentiability-II to see how the term is derived!

Setting $h := x - a$ and $k := y - b$, we get that

$$\begin{aligned} \mathcal{R}((x, y), (a, b)) &= \frac{1}{2} \left[f_{xx}(\xi, \eta)h^2 + 2f_{xy}(\xi, \eta)hk + f_{yy}(\xi, \eta)k^2 \right] \\ &= \frac{1}{2}k^2 \left[f_{xx}(\xi, \eta)t^2 + 2f_{xy}(\xi, \eta)t + f_{yy}(\xi, \eta) \right] \quad \text{for } t = \frac{h}{k} \end{aligned}$$

Since $\frac{1}{2}k^2 > 0$, the quantity $\mathcal{R}((x, y), (a, b))$ has the same sign as the expression inside the square brackets

$$\underbrace{f_{xx}(\xi, \eta)t^2 + 2f_{xy}(\xi, \eta)t + f_{yy}(\xi, \eta)}_{=: P(t)}$$

which is a quadratic function $P(t)$ whose sign is determined by the signs of

$$\underbrace{\Delta(\xi, \eta) := [f_{xy}(\xi, \eta)]^2 - f_{xx}(\xi, \eta)f_{yy}(\xi, \eta)}_{\text{the discriminant}} \quad \text{and} \quad f_{xx}(\xi, \eta) \quad \text{or} \quad f_{yy}(\xi, \eta) \quad (*) .$$

For instance, if $\Delta(\xi, \eta) < 0$ and $f_{xx}(\xi, \eta) > 0$, then $P(t) > 0$ and hence, $\mathcal{R}((a, b), (x, y)) > 0$ which implies that (a, b) is a point of local minimum for the function (a, b) .

Once again, let us bear in mind that we want to determine whether the critical point (a, b) is a point of local max/min or neither. So, the running points (x, y) are taken in a small disc centered at (a, b) . Since (ξ, η) lies on the segment line joining (a, b) to (x, y) , it follows that also (ξ, η) lies in that small disc.

From the continuity of the second order partial derivatives $f_{xx}, f_{xy} = f_{yx}, f_{yy}$ it is sufficient to obtain the information required in the box $(*)$ above, from the signs of the following terms:

$$\Delta(a, b) := [f_{xy}(a, b)]^2 - f_{xx}(a, b)f_{yy}(a, b) \quad \text{and} \quad f_{xx}(a, b) \quad \text{or} \quad f_{yy}(a, b) \quad (1)$$

The following important result gives a sets of sufficient conditions of optimality.

Theorem 0.5 (A sufficient condition of optimality)

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function of class $\mathcal{C}^2$ and let (a, b) be a critical point for the function f . Set

$$D(a, b) := \begin{vmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{xy}(a, b) & f_{yy}(a, b) \end{vmatrix} = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2 = -\Delta(a, b) \quad (\text{see (1)}) .$$

Notice that $D(a, b)$ is the determinant of the Hessian matrix $Hf(a, b)$ defined by

$$Hf(a, b) = \begin{pmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{xy}(a, b) & f_{yy}(a, b) \end{pmatrix}.$$

- (i) If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$ or $f_{yy}(a, b) > 0$, then the critical point (a, b) is a point of local minimum for the function f .
- (ii) If $D(a, b) > 0$ and $f_{xx}(a, b) < 0$ or $f_{yy}(a, b) < 0$, then the critical point (a, b) is a point of local maximum for the function f .
- (iii) If $D(a, b) < 0$, then the critical point (a, b) is neither a point of local minimum nor maximum for the function f . The critical point (a, b) is called in this case a **saddle point** for the function f .
- (iv) If $D(a, b) = 0$ then we have an undeterminate case for which anything can happen.

Remark 0.6 We would like to illustrate through examples what we mean by “anything can happen”.

Consider the following three functions:

$$f_1(x, y) := x^4 + y^4, \quad f_2(x, y) := -x^4 - y^4, \quad f_3(x, y) := x^4 - y^4. \quad (2)$$

The point $(0, 0)$ is a critical point for all the three functions and also

$$Hf_1(0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad Hf_2(0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad Hf_3(0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Therefore, the method described above in Theorem 0.5 does not allow us to determine the nature of the point $(0, 0)$ for each of these functions. However,

$$\begin{aligned} f_1(x, y) &\geq 0 = f_1(0, 0) &\Rightarrow & (0, 0) \text{ is a point of global minimum for } f_1, \\ f_2(x, y) &\leq 0 = f_2(0, 0) &\Rightarrow & (0, 0) \text{ is a point of global maximum for } f_2, \\ \left. \begin{aligned} f_3(x, 0) &\geq 0 = f_3(0, 0) \\ f_3(0, y) &\leq 0 = f_3(0, 0) \end{aligned} \right\} &\Rightarrow & (0, 0) \text{ is a saddle point for } f_3. \end{aligned}$$

INDEED, ANYTHING CAN HAPPEN!

Example 0.7 Consider the function $f(x, y) := (x + y)^2 - (x + 5y + xy)$.

- (a) Find all the critical points for the function f .
- (b) Determine the nature of each critical point found in (a).

Solution:

- (a) We want to find (x, y) such that

$$\begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases} \Leftrightarrow \begin{cases} 2x + y = 1 \\ x + 2y = 5 \end{cases} \Leftrightarrow \Leftrightarrow x = -1, \quad y = 3.$$

So, the point $(-1, 3)$ is the only critical point for the function f .

- (b) Now, we need to check whether the point $(-1, 3)$ satisfies the second order conditions in Theorem 0.5. We will need

$$f_{xx}(x, y) = 2, \quad f_{xy}(x, y) = 1, \quad f_{yy} = 2.$$

So,

$$D(-1, 3) = \begin{vmatrix} f_{xx}(-1, 3) & f_{xy}(-1, 3) \\ f_{xy}(-1, 3) & f_{yy}(-1, 3) \end{vmatrix} = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 3 > 0 \text{ while } f_{xx}(-1, 3) = 2 > 0.$$

Therefore $(-1, 3)$ is a point of local minimum for the function f .

Example 0.8 Consider the function $f(x, y) := x^2 + xy^3 + x$.

- (a) Find all the critical points for the function f .
 (b) Determine the nature of each critical point found in (a).

Solution:

- (a) We want to find (x, y) such that

$$\begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases} \Leftrightarrow \begin{cases} 2x + y^3 + 1 = 0 \\ 3xy^2 = 0 \end{cases}.$$

From the second equation, we find that $x = 0$ or $y = 0$.

If $x = 0$, we get from the first equation that $y = -1$ and hence we obtain the point $(0, -1)$.

If $y = 0$, we get from the second equation that $x = -\frac{1}{2}$ and hence, we obtain the point $(-1/2, 0)$.

So the critical points are

$$(0, -1) \quad \text{and} \quad (-1/2, 0).$$

- (b) (i) We need to check whether the point $(0, -1)$ satisfies the second order conditions in Theorem 0.5. We will need

$$f_{xx}(x, y) = 2, \quad f_{xy}(x, y) = 3y^2, \quad f_{yy} = 6xy.$$

So,

$$D(0, -1) = \begin{vmatrix} f_{xx}(0, -1) & f_{xy}(0, -1) \\ f_{xy}(0, -1) & f_{yy}(0, -1) \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ 3 & 0 \end{vmatrix} = -9 < 0.$$

Thus, $(0, -1)$ is a saddle point.

- (ii) We need to check whether the point $(-1/2, 0)$ satisfies the second order conditions in Theorem 0.5. So,

$$D(-1/2, 0) = \begin{vmatrix} f_{xx}(-1/2, 0) & f_{xy}(-1/2, 0) \\ f_{xy}(-1/2, 0) & f_{yy}(-1/2, 0) \end{vmatrix} = \begin{vmatrix} 2 & 0 \\ 0 & 0 \end{vmatrix} = 0 \quad (\text{Undeterminate}).$$

In this case, you need to analyse directly the function f . For instance, taking $x = -1/2$ in the expression of f , we obtain the function of the single variable y defined by

$$f(-1/2, y) = -\frac{1}{2}y^3 - \frac{1}{4} \quad \text{for which } f_y(-1/2, y) = -\frac{3}{2}y^2 \leq 0.$$

Hence, the function has neither local maximum nor minimum at $y = 0$. Therefore, $(-1/2, 0)$ is also a saddle point.

We could also notice that taking $y = 0$ in the expression of f , we obtain the function of the single variable x defined by $f(x, 0) = x^2 + x$ which has a global minimum at $x = -\frac{1}{2}$.

Example 0.9 Consider the function $f(x, y) := y^2 + x^2y + x^4$.

- (a) Find all the critical points for the function f .
- (b) Determine the nature of each critical point found in (a). [Hint: Notice from square completion that $f(x, y) = x^4 + x^2y + y^2 = (x^2 + \frac{1}{2}y)^2 + \frac{3}{4}y^2 \geq 0$].

Solution:

- (a) We want to find (x, y) such that

$$\begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases} \Leftrightarrow \begin{cases} 2xy + 4x^3 = 0 \\ 2y + x^2 = 0 \end{cases} \Leftrightarrow \begin{cases} 2x(y + 2x^2) = 0 \\ 2y + x^2 = 0 \end{cases} \Leftrightarrow x = y = 0.$$

So, the point $(0, 0)$ is the only critical point for the function f .

- (b) Now, we need to check whether the point $(0, 0)$ satisfies the second order conditions in Theorem 0.5. We will need

$$f_{xx}(x, y) = 2y + 12x^2, \quad f_{xy}(x, y) = 2x, \quad f_{yy} = 2.$$

So,

$$D(0, 0) = \begin{vmatrix} f_{xx}(0, 0) & f_{xy}(0, 0) \\ f_{xy}(0, 0) & f_{yy}(0, 0) \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ 0 & 2 \end{vmatrix} = 0 \quad (\text{Undetermined case!}).$$

However, since $f(x, y) \geq 0 = f(0, 0)$, we get that $(0, 0)$ is a point of global minimum for the function f according to (b) in Definition 0.2.

Exercises 0.10 Find the points of local/global maximum/minimum, saddle points for each of the following functions:

- (a) $f_1(x, y) := x^3 + y^3 - 3axy$ for $a \geq 0$;
- (b) $f_2(x, y) := xy^2(5 - x - y)^2$;
- (c) $f_3(x, y) := (x - 2)^4 + (x - y)^4$;
- (d) $f_4(x, y) := \cos(x + y) + \cos(x - y)$;
- (e) $f_5(x, y) := x(x + y)e^{x-y}$;
- (f) $f_6(x, y) := xe^y + ye^x$.

Hint: For (f) you will need to show that the equation $e^x + xe^{\frac{1}{x}} = 0$ has the unique solution $x = -1$.

The next set of notes will be on finding maximum and minimum subject to one equality constraint!